Atmospheric entry

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 7.*
$\Delta U = +20$ J.

02. Worked example: explain the model
Branches 700 and 300 J; could measure temperature change, emitted radiation or air heating with appropriate models.

03. Read and interpret the model
Smaller sphere has greater area per mass; stronger surface exposure per kg, all else equal.

04. Apply the physics
Ratio 5; no, energy partitions among air, radiation, deformation and the object.

05. Explain your reasoning
Entry angle/speed profile, density/strength, fragmentation, shape, heat-transfer rate, atmosphere; accept relevant factors.

06. Review: connect an earlier idea *Review: Unit 4, Day 20.*
No; it lacks 4 J of the required energy and remains bound under the model.

07. Exit check
Larger has half area per mass.

08. STRETCH: test the model
200 and 600 J; need mass, material, shape, heat transfer, ablation and changing conditions.

Atmospheric entry

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 7.*
$Q=\Delta U+W_{by}=6$ J; heat enters.

02. Worked example: explain the model
Branches 700 and 300 J; could measure temperature change, emitted radiation or air heating with appropriate models.

03. Read and interpret the model
Smaller sphere has greater area per mass; stronger surface exposure per kg, all else equal.

04. Apply the physics
K 1.125×10^{9} J; requirement 8.6×10^{7} J; ratio 13.1. This does not determine survival.

05. Explain your reasoning
Entry angle/speed profile, density/strength, fragmentation, shape, heat-transfer rate, atmosphere; accept relevant factors.

06. Review: connect an earlier idea *Review: Unit 4, Day 20.*
New speed is 1/4; $v_{escape}=\sqrt{(2\ GM/r)}$.

07. Exit check
$A/m\propto1/r$, so half. KE per mass depends on speed; survival also depends on transfer rates, fragmentation and conditions.

08. STRETCH: test the model
$\Delta T=20$ K; simplified lumped model omits gradients, ablation, radiation and material failure.